

Principal Component Analysis and Singular Value Decomposition: A Comprehensive Survey for Advanced Predictive Modeling

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Abstract—Principal Component Analysis (PCA) and Singular Value Decomposition (SVD) are two of the most foundational and widely deployed techniques in modern data science and machine learning. Despite their distinct mathematical origins, PCA and SVD are intimately connected: PCA is most efficiently computed through SVD, and both methods serve as cornerstones for dimensionality reduction, latent structure discovery, and noise filtering in high-dimensional datasets. This survey provides a rigorous and self-contained treatment of both methods, covering their mathematical underpinnings, algorithmic implementations, and practical applications within the context of advanced predictive modeling. We examine how these techniques expose linear dependencies in structured data, enable the construction of lower-dimensional representations that preserve maximum variance, and support downstream prediction tasks by accounting for correlation structure. We further survey their application in areas including genomics, image compression, natural language processing, collaborative filtering, and time-series analysis. Connections to related methods such as factor analysis, kernel PCA, independent component analysis, and truncated SVD are explored. Limitations, assumptions, and practical considerations for numerical stability are also discussed. This survey synthesizes the extensive existing literature and is intended as a comprehensive reference for graduate students and practitioners in data science.

Index Terms—principal component analysis, singular value decomposition, dimensionality reduction, latent structure, eigendecomposition, low-rank approximation, predictive modeling

I. Introduction

The explosion of high-dimensional data in modern science and industry has made dimensionality reduction an essential preprocessing step for virtually every predictive modeling pipeline. When features number in the thousands or millions, naïve regression and classification approaches suffer from the curse of dimensionality: distances become uninformative, covariance estimates become singular, and overfitting becomes severe [23]. The need for methods that extract parsimonious, interpretable

representations from complex data has therefore never been greater.

Principal Component Analysis (PCA) and Singular Value Decomposition (SVD) occupy a privileged position in this landscape. PCA, introduced by Pearson in 1901 [1] and independently developed by Hotelling in 1933 [2], finds a set of orthogonal directions in feature space—the principal components—along which the data variance is maximized. SVD, whose theoretical roots trace to the work of Beltrami, Jordan, and Sylvester in the 1870s and to modern algorithmic developments by Golub and Kahan [3], decomposes any matrix into a product of three structured matrices, revealing the intrinsic geometry of linear maps and data matrices alike.

Both methods address several interrelated challenges in data science. First, they identify dependencies in structured data: correlation among features is precisely what PCA exploits to reduce dimensionality without losing information. Second, they provide a principled basis for model construction in the presence of dependent inputs, replacing correlated predictors with uncorrelated components. Third, they support forecasting and prediction by concentrating signal in a small number of components and discarding noise. Fourth, and perhaps most profoundly, they reveal latent structure—the hidden low-dimensional geometry that underlies complex, high-dimensional observations.

This survey is organized as follows. Section II establishes mathematical notation and foundational concepts from linear algebra and statistics. Section III develops the core theory of SVD and PCA, including the Eckart–Young–Mirsky theorem and the relationship between the two methods. Section IV describes practical algorithms for computation. Section V presents worked numerical examples. Section VI surveys applications across several domains of data science. Section VII discusses assumptions

and limitations. Section VIII connects PCA and SVD to related methods. Section IX concludes.

II. Mathematical Preliminaries

A. Notation

Throughout this paper, we denote scalars by lowercase italic letters (a, b, λ), column vectors by bold lowercase letters ($\mathbf{x}, \mathbf{v}$), and matrices by bold uppercase letters ($\mathbf{A}, \mathbf{X}, \mathbf{\Sigma}$). The transpose of matrix $\mathbf{A}$ is $\mathbf{A}^\top$, and its Moore–Penrose pseudoinverse is $\mathbf{A}^+$. The Frobenius norm is $\|\mathbf{A}\|_F = \sqrt{\sum_{i,j} a_{ij}^2}$, and the spectral (operator) norm is $\|\mathbf{A}\|_2 = \sigma_1(\mathbf{A})$, the largest singular value. $\mathbf{I}_n$ denotes the $n \times n$ identity matrix. $\mathbb{R}^{n \times p}$ is the space of real $n \times p$ matrices.

B. Eigendecomposition

Definition 1 (Eigenvalue Decomposition). A square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$ admits an eigendecomposition $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^{-1}$ if there exist a diagonal matrix $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_n)$ and an invertible matrix $\mathbf{Q}$ whose columns are the corresponding eigenvectors $\mathbf{q}_i$ satisfying $\mathbf{A}\mathbf{q}_i = \lambda_i\mathbf{q}_i$.

For symmetric positive semidefinite (SPSD) matrices—such as sample covariance matrices—the eigendecomposition becomes the spectral decomposition: $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ with orthonormal eigenvectors $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}$ and non-negative eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n \geq 0$.

C. Sample Covariance Matrix

Given a data matrix $\mathbf{X} \in \mathbb{R}^{n \times p}$ with n observations and p features, let $\bar{\mathbf{x}} = \frac{1}{n}\mathbf{X}^\top\mathbf{1}_n$ denote the sample mean vector. The centered data matrix is $\tilde{\mathbf{X}} = \mathbf{X} - \mathbf{1}_n\bar{\mathbf{x}}^\top$. The sample covariance matrix is:

$$\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} \in \mathbb{R}^{p \times p}. \quad (1)$$

$\mathbf{S}$ is symmetric positive semidefinite, with $\text{rank}(\mathbf{S}) \leq \min(n-1, p)$.

D. Variance and Covariance Structure

The (i, j) entry of $\mathbf{S}$ is $s_{ij} = \frac{1}{n-1} \sum_{k=1}^n \tilde{x}_{ki}\tilde{x}_{kj}$, measuring the linear dependence between features i and j . When $\mathbf{S}$ has many non-zero off-diagonal entries, the features are mutually dependent, and predictive models trained on raw features may be poorly conditioned. PCA addresses this by finding a basis in which the covariance matrix is diagonal.

III. Core Theory

A. Singular Value Decomposition

Theorem 1 (Existence and Uniqueness of SVD [4]). Every matrix $\mathbf{A} \in \mathbb{R}^{n \times p}$ with $n \geq p$ admits a singular value decomposition

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top, \quad (2)$$

where $\mathbf{U} \in \mathbb{R}^{n \times n}$ and $\mathbf{V} \in \mathbb{R}^{p \times p}$ are orthogonal matrices ($\mathbf{U}^\top\mathbf{U} = \mathbf{I}_n$, $\mathbf{V}^\top\mathbf{V} = \mathbf{I}_p$), and $\mathbf{\Sigma} \in \mathbb{R}^{n \times p}$ is a rectangular diagonal matrix with non-negative entries $\sigma_1 \geq \sigma_2 \geq \dots \geq$

$\sigma_r > 0 = \sigma_{r+1} = \dots$, where $r = \text{rank}(\mathbf{A})$. The values σ_i are called the singular values of $\mathbf{A}$, the columns $\mathbf{u}_i$ of $\mathbf{U}$ are the left singular vectors, and the columns $\mathbf{v}_i$ of $\mathbf{V}$ are the right singular vectors. The singular values are unique; the singular vectors are unique when all singular values are distinct.

The thin (economy) SVD retains only the r non-trivial factors: $\mathbf{A} = \mathbf{U}_r\mathbf{\Sigma}_r\mathbf{V}_r^\top$, where $\mathbf{U}_r \in \mathbb{R}^{n \times r}$, $\mathbf{\Sigma}_r \in \mathbb{R}^{r \times r}$, $\mathbf{V}_r \in \mathbb{R}^{p \times r}$.

The SVD can be understood as an outer product expansion:

$$\mathbf{A} = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \quad (3)$$

revealing that every rank- r matrix is a sum of r rank-1 matrices weighted by the singular values.

B. The Eckart–Young–Mirsky Theorem

A fundamental consequence of SVD is the optimal low-rank approximation theorem.

Theorem 2 (Eckart–Young–Mirsky [5]). For any $k \leq r$, the rank- k matrix that minimizes the Frobenius norm approximation error is

$$\mathbf{A}_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^\top = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top, \quad (4)$$

and the error is

$$\|\mathbf{A} - \mathbf{A}_k\|_F^2 = \sum_{i=k+1}^r \sigma_i^2. \quad (5)$$

The same truncation is also optimal in the spectral norm: $\|\mathbf{A} - \mathbf{A}_k\|_2 = \sigma_{k+1}$.

This theorem is the theoretical justification for dimensionality reduction: if the singular values decay rapidly, a small k captures most of the information in $\mathbf{A}$.

C. Principal Component Analysis

Definition 2 (Principal Components). Let $\tilde{\mathbf{X}} \in \mathbb{R}^{n \times p}$ be mean-centered data. The principal components of $\tilde{\mathbf{X}}$ are the orthonormal directions $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_p \in \mathbb{R}^p$ defined sequentially by

$$\mathbf{w}_j = \arg \max_{\|\mathbf{w}\|=1, \mathbf{w} \perp \mathbf{w}_1, \dots, \mathbf{w}_{j-1}} \mathbf{w}^\top \mathbf{S} \mathbf{w}, \quad (6)$$

where $\mathbf{S}$ is the sample covariance matrix.

Theorem 3 (PCA via Spectral Decomposition). The principal components $\mathbf{w}_j$ are the eigenvectors of the sample covariance matrix $\mathbf{S}$, ordered by decreasing eigenvalue:

$$\mathbf{S} = \mathbf{W}\mathbf{\Lambda}\mathbf{W}^\top, \quad \mathbf{W} = [\mathbf{w}_1 | \dots | \mathbf{w}_p], \quad (7)$$

with $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$. The variance of the j -th principal component score $z_j = \tilde{\mathbf{X}}\mathbf{w}_j$ equals λ_j .

The principal component scores (projected data) are collected into the matrix:

$$\mathbf{Z} = \tilde{\mathbf{X}}\mathbf{W} \in \mathbb{R}^{n \times p}, \quad (8)$$

and the rank- k PCA reconstruction of the centered data is:

$$\hat{\tilde{\mathbf{X}}}_k = \mathbf{Z}_k \mathbf{W}_k^\top = \tilde{\mathbf{X}}\mathbf{W}_k \mathbf{W}_k^\top, \quad (9)$$

where $\mathbf{W}_k$ contains the leading k eigenvectors.

D. The PCA–SVD Connection

The intimate relationship between PCA and SVD is made precise by the following result [6].

Proposition 1 (PCA from SVD). Let $\tilde{\mathbf{X}} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ be the SVD of the centered data matrix. Then:

- 1) The right singular vectors $\mathbf{V} = \mathbf{W}$ are the principal component directions (eigenvectors of $\mathbf{S}$).
- 2) The singular values satisfy $\sigma_j = \sqrt{(n-1)\lambda_j}$, connecting them to the eigenvalues of $\mathbf{S}$.
- 3) The principal component scores are $\mathbf{Z} = \mathbf{U}\mathbf{\Sigma}$, the left singular vectors scaled by singular values.

Proof. By definition, $\mathbf{S} = \frac{1}{n-1}\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}}$. Substituting the SVD:

$$\begin{aligned} \mathbf{S} &= \frac{1}{n-1}(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top)^\top(\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top) \\ &= \frac{1}{n-1}\mathbf{V}\mathbf{\Sigma}^\top\mathbf{U}^\top\mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \\ &= \mathbf{V}\left(\frac{\mathbf{\Sigma}^2}{n-1}\right)\mathbf{V}^\top. \end{aligned} \quad (10)$$

This is the spectral decomposition of $\mathbf{S}$, confirming $\mathbf{W} = \mathbf{V}$ and $\lambda_j = \sigma_j^2/(n-1)$. $\square$

This equivalence means PCA can be computed without ever forming the $p \times p$ covariance matrix—an important practical advantage when p is large and $n \ll p$.

E. Proportion of Variance Explained

A key diagnostic for choosing the number of components k is the proportion of variance explained (PVE):

$$\text{PVE}(k) = \frac{\sum_{j=1}^k \lambda_j}{\sum_{j=1}^p \lambda_j} = \frac{\sum_{j=1}^k \sigma_j^2}{\sum_{j=1}^p \sigma_j^2} = \frac{\|\mathbf{A}_k\|_F^2}{\|\mathbf{A}\|_F^2}. \quad (11)$$

A scree plot displays λ_j against j ; the “elbow” heuristic selects k at the point of diminishing returns. Alternatively, one selects the smallest k such that $\text{PVE}(k) \geq \tau$ for a threshold τ (commonly 0.85 or 0.95).

IV. Algorithms and Methods

A. Full SVD via Golub–Reinsch Bidiagonalization

The standard algorithm for computing the full SVD of a dense matrix $\mathbf{A} \in \mathbb{R}^{n \times p}$ proceeds in two phases [3], [4].

Algorithm 1: Golub–Reinsch SVD

- 1) Bidiagonalization. Apply a sequence of Householder reflections from the left and right to reduce $\mathbf{A}$ to an upper bidiagonal matrix $\mathbf{B}$:

$$\mathbf{U}_1^\top \mathbf{A} \mathbf{V}_1 = \mathbf{B} = \begin{pmatrix} d_1 & f_1 & & \\ & d_2 & \ddots & \\ & & \ddots & f_{p-1} \\ & & & d_p \end{pmatrix}. \quad (12)$$

This requires $O(np^2)$ operations.

- 2) Diagonalization. Apply the QR algorithm with implicit shifts to $\mathbf{B}$, accumulating orthogonal transformations, until $\mathbf{B}$ converges to diagonal form. This yields $\mathbf{U}_2^\top \mathbf{B} \mathbf{V}_2 = \mathbf{\Sigma}$.
- 3) Assembly. Set $\mathbf{U} = \mathbf{U}_1 \mathbf{U}_2$ and $\mathbf{V} = \mathbf{V}_1 \mathbf{V}_2$.

Total cost: $O(np^2 + p^3)$ for $n \geq p$. Numerical stability is guaranteed by the use of unitary transformations [7].

B. Truncated SVD via Randomized Methods

For large-scale problems, computing only the leading k singular triplets is sufficient. The randomized SVD algorithm of Halko, Martinsson, and Tropp [8] provides a highly efficient approach.

Algorithm 2: Randomized SVD

- 1) Random projection. Draw a random Gaussian matrix $\mathbf{\Omega} \in \mathbb{R}^{p \times (k+s)}$ where s is a small oversampling parameter (e.g., $s = 10$). Compute $\mathbf{Y} = \mathbf{A}\mathbf{\Omega} \in \mathbb{R}^{n \times (k+s)}$.
- 2) (Optional) Power iteration. Replace $\mathbf{Y} \leftarrow (\mathbf{A}\mathbf{A}^\top)^q \mathbf{Y}$ for $q \geq 1$ to improve approximation for slowly decaying spectra.
- 3) QR factorization. Compute $\mathbf{Y} = \mathbf{Q}\mathbf{R}$ via QR decomposition to obtain an orthonormal basis $\mathbf{Q} \in \mathbb{R}^{n \times (k+s)}$ for the column space of $\mathbf{A}$.
- 4) Projection. Form the small matrix $\mathbf{B} = \mathbf{Q}^\top \mathbf{A} \in \mathbb{R}^{(k+s) \times p}$.
- 5) SVD of small matrix. Compute $\mathbf{B} = \hat{\mathbf{U}}\mathbf{\Sigma}\mathbf{V}^\top$ exactly.
- 6) Recover left singular vectors. Set $\mathbf{U} = \mathbf{Q}\hat{\mathbf{U}}$ and truncate to the leading k components.

Total cost: $O(nk \log k)$ matrix-vector products with $\mathbf{A}$, far cheaper than the full SVD for $k \ll \min(n, p)$. This algorithm is implemented in `sklearn.utils.extmath.randomized_svd` [9].

C. Incremental PCA for Streaming Data

When data arrive in a stream or do not fit in memory, incremental PCA [10] updates the principal components with each new batch $\mathbf{X}_t$ without retaining the full history:

$$\mathbf{S}_t = \frac{(n_t - 1)\mathbf{S}_{t-1} + (n_{\text{new}} - 1)\mathbf{S}_{\text{new}} + \delta_t}{n_t + n_{\text{new}} - 1}, \quad (13)$$

where δ_t is a correction term for the shift in means. The eigenvectors of $\mathbf{S}_t$ are updated via a rank- k SVD of the augmented matrix formed from the previous basis and new data.

TABLE I
Bivariate Dataset for PCA Illustration

Observation	x_1	x_2
1	2.5	2.4
2	0.5	0.7
3	2.2	2.9
4	1.9	2.2
5	3.1	3.0
Mean	2.04	2.24

D. Choosing the Number of Components

Several criteria guide the selection of k :

- Scree plot elbow: select k at the “knee” of the scree plot.
- Cumulative PVE threshold: smallest k with $\text{PVE}(k) \geq 0.90$.
- Parallel analysis [11]: retain components whose eigenvalues exceed those from random data.
- Broken stick model [12]: compare eigenvalues to the expected distribution under a null model.
- Cross-validation: minimize reconstruction error on held-out data [13].

V. Worked Examples

A. Small-Scale SVD Computation

Consider the matrix

$$\mathbf{A} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \\ 0 & 0 \end{pmatrix} \in \mathbb{R}^{3 \times 2}. \quad (14)$$

We compute $\mathbf{A}^\top \mathbf{A} = \begin{pmatrix} 10 & 6 \\ 6 & 10 \end{pmatrix}$, whose eigenvalues are $\lambda_1 = 16$, $\lambda_2 = 4$. The singular values are $\sigma_1 = 4$, $\sigma_2 = 2$.

Eigenvectors of $\mathbf{A}^\top \mathbf{A}$: $\mathbf{v}_1 = \frac{1}{\sqrt{2}}(1, 1)^\top$, $\mathbf{v}_2 = \frac{1}{\sqrt{2}}(1, -1)^\top$.

Left singular vectors: $\mathbf{u}_i = \frac{1}{\sigma_i} \mathbf{A} \mathbf{v}_i$:

$$\mathbf{u}_1 = \frac{1}{4} \begin{pmatrix} 3 & 1 \\ 1 & 3 \\ 0 & 0 \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{4\sqrt{2}} \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad (15)$$

$$\mathbf{u}_2 = \frac{1}{2\sqrt{2}} \begin{pmatrix} 3 & -1 \\ 1 & -3 \\ 0 & 0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}. \quad (16)$$

The third left singular vector $\mathbf{u}_3 = (0, 0, 1)^\top$ spans the null space. The full SVD is:

$$\mathbf{A} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 4 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & 0 \end{pmatrix}. \quad (17)$$

TABLE II
Low-Rank Approximation Error and Cumulative Variance Explained

Rank k	σ_k	$\ \mathbf{A} - \mathbf{A}_k\ _F^2$	PVE (%)
0	—	154.50	0.0
1	10.00	54.50	64.7
2	6.00	18.50	88.0
3	3.00	9.25	94.0
4	1.50	7.00	95.5
5	0.50	0.00	100.0

B. PCA on Correlated Bivariate Data

Suppose we observe $n = 5$ data points with $p = 2$ features shown in Table I.

After centering, the sample covariance matrix is approximately:

$$\mathbf{S} \approx \begin{pmatrix} 0.616 & 0.615 \\ 0.615 & 0.716 \end{pmatrix}. \quad (18)$$

The eigenvalues are $\lambda_1 \approx 1.284$ and $\lambda_2 \approx 0.049$, with eigenvectors $\mathbf{w}_1 \approx (0.677, 0.736)^\top$ and $\mathbf{w}_2 \approx (-0.736, 0.677)^\top$. The first principal component explains $\text{PVE}(1) = 1.284/1.333 \approx 96.3\%$ of the variance, confirming that the data lie nearly on a one-dimensional manifold. This strong first component reflects the high linear correlation ($r \approx 0.926$) between features x_1 and x_2 .

C. Low-Rank Approximation Quality

Table II illustrates the Eckart–Young theorem for a hypothetical rank-5 matrix with singular values $\sigma = (10, 6, 3, 1.5, 0.5)$.

VI. Applications in Data Science

A. Genomics and Gene Expression Analysis

High-throughput sequencing produces data matrices with n samples and $p \sim 10^4$ – 10^5 gene expression features. PCA is a standard first step in exploratory analysis of such data [14]: the leading principal components often capture population structure, batch effects, or major biological variation. In genome-wide association studies (GWAS), including the top PCs as covariates in regression models corrects for population stratification and reduces spurious associations [15]. SVD is also used in RNA-seq analysis for deconvolution of cell-type mixtures [16].

B. Image Compression and Reconstruction

A grayscale image can be represented as a matrix $\mathbf{A} \in \mathbb{R}^{n \times p}$ where each entry is a pixel intensity. The truncated SVD $\mathbf{A}_k$ provides a compact representation requiring storage of $k(n + p + 1)$ values rather than np . For a 512×512 image with $r = 512$, retaining $k = 50$ singular components (roughly 10% of the components) typically yields visually acceptable reconstruction with $\|\mathbf{A} - \mathbf{A}_{50}\|_F / \|\mathbf{A}\|_F < 0.05$ [37]. This demonstrates SVD as a lossy compression mechanism governed by the Eckart–Young theorem.

C. Natural Language Processing: Latent Semantic Analysis

Latent Semantic Analysis (LSA), introduced by Deerwester et al. [17], applies truncated SVD to a term-document matrix $\mathbf{A}$ (often with TF-IDF weighting): $\mathbf{A}_k = \mathbf{U}_k \mathbf{\Sigma}_k \mathbf{V}_k^\top$. The rows of $\mathbf{U}_k \mathbf{\Sigma}_k$ represent documents in a k -dimensional semantic space, and cosine similarity in this space captures semantic relatedness better than raw word overlap. LSA identifies latent topics as linear combinations of terms and served as an early precursor to probabilistic topic models [18].

D. Recommender Systems: Collaborative Filtering

In collaborative filtering, the user-item rating matrix $\mathbf{R} \in \mathbb{R}^{m \times n}$ is typically sparse with millions of missing entries. The Netflix Prize competition popularized matrix factorization [19]: find low-rank matrices $\mathbf{P} \in \mathbb{R}^{m \times k}$ (user factors) and $\mathbf{Q} \in \mathbb{R}^{n \times k}$ (item factors) minimizing

$$\min_{\mathbf{P}, \mathbf{Q}} \sum_{(i,j) \in \Omega} (r_{ij} - \mathbf{p}_i^\top \mathbf{q}_j)^2 + \lambda (\|\mathbf{P}\|_F^2 + \|\mathbf{Q}\|_F^2), \quad (19)$$

where Ω is the set of observed ratings. This is a generalization of SVD to incomplete matrices, solved by alternating least squares or stochastic gradient descent. The latent factors $\mathbf{p}_i$ and $\mathbf{q}_j$ capture user preferences and item attributes implicitly.

E. Multivariate Time Series and Forecasting

PCA applied to multivariate time series extracts the dominant modes of variation—often corresponding to interpretable economic, climatic, or financial factors. In macroeconomic forecasting, the diffusion index approach of Stock and Watson [20] estimates common factors via PCA from a large panel of economic indicators and uses them as predictors in autoregressive models:

$$y_{t+h} = \alpha + \sum_{j=1}^k \beta_j \hat{f}_{jt} + \sum_{l=1}^L \gamma_l y_{t-l+1} + \epsilon_{t+h}, \quad (20)$$

where $\hat{f}_{jt}$ are PCA-estimated factors at time t . This framework explicitly accounts for dependence structure across variables and time, improving forecast accuracy over univariate models.

F. Signal Processing: Empirical Orthogonal Functions

In climatology and oceanography, PCA applied to spatiotemporal data fields yields empirical orthogonal functions (EOFs) [21]. Each EOF $\mathbf{w}_j$ describes a spatial pattern, and the corresponding principal component time series $z_j(t)$ describes how that pattern evolves over time. The first EOF of sea surface temperature, for example, often captures the El Niño–Southern Oscillation (ENSO) signal—a dominant mode of interannual climate variability. SVD applied to cross-covariance matrices of two fields yields maximum covariance analysis (MCA), identifying coupled patterns of variability.

G. Dimensionality Reduction for Predictive Models

In high-dimensional settings where $p \gg n$, direct regression of a response $\mathbf{y}$ on features $\mathbf{X}$ is infeasible or highly regularized. Principal Component Regression (PCR) [38] regresses $\mathbf{y}$ on the leading k PCA scores:

$$\hat{\mathbf{y}} = \mathbf{Z}_k \hat{\boldsymbol{\beta}}_k, \quad \hat{\boldsymbol{\beta}}_k = (\mathbf{Z}_k^\top \mathbf{Z}_k)^{-1} \mathbf{Z}_k^\top \mathbf{y}. \quad (21)$$

Because the columns of $\mathbf{Z}_k$ are orthogonal, $\mathbf{Z}_k^\top \mathbf{Z}_k = \mathbf{I}_k$ is diagonal, rendering the regression perfectly conditioned. Substituting the PCA decomposition, the predictions in the original feature space are:

$$\hat{\mathbf{y}} = \mathbf{X} \mathbf{W}_k (\mathbf{W}_k^\top \mathbf{X}^\top \mathbf{X} \mathbf{W}_k)^{-1} \mathbf{W}_k^\top \mathbf{X}^\top \mathbf{y}. \quad (22)$$

PCR is closely related to ridge regression [22] and partial least squares (PLS), with PCR selecting components to maximize variance in $\mathbf{X}$ and PLS selecting components to maximize covariance with $\mathbf{y}$.

VII. Limitations and Assumptions

A. Linearity

PCA and SVD are inherently linear methods: they decompose the data into linear combinations of basis vectors. If the true latent structure is nonlinear (e.g., data lying on a curved manifold), PCA may require many components to achieve an adequate approximation. Kernel PCA [24] and manifold learning methods (Isomap, LLE, UMAP [36]) address this limitation.

B. Gaussian and Second-Order Assumptions

PCA maximizes variance, a second-order statistic. It is the optimal linear representation for multivariate Gaussian data, but when data are non-Gaussian, the principal components are not guaranteed to be statistically independent—only uncorrelated. Independent Component Analysis (ICA) [25] seeks components that are statistically independent, exploiting higher-order statistics.

C. Scale Sensitivity

PCA is sensitive to the units of measurement. Features measured on different scales will contribute disproportionately to the total variance. Standard practice is to standardize features to unit variance before applying PCA (yielding correlation PCA), so that all variables contribute equally. However, standardization discards information about the relative variability of features and may not always be appropriate.

D. Missing Data

Standard PCA and SVD require complete data matrices. In practice, data often contain missing values. Several approaches exist: imputation prior to PCA, the EM algorithm for PCA [27], probabilistic PCA [26], and nuclear norm minimization for matrix completion [28]. Each carries additional assumptions about the missing data mechanism.

E. Interpretability of Components

Principal components are linear combinations of all original features and need not correspond to any physically interpretable quantity. Rotation methods (Varimax, Oblimin) in factor analysis [29] improve interpretability by producing sparser loadings, at the cost of losing orthogonality. Sparse PCA [30] directly constrains loading vectors to be sparse via ℓ_1 penalties.

F. Computational Considerations

For $n \gg p$, the covariance matrix approach (eigen-decomposition of $\mathbf{S} \in \mathbb{R}^{p \times p}$) is more efficient: $O(np^2 + p^3)$. For $p \gg n$, SVD of $\mathbf{X}$ directly is more efficient: $O(n^2p + n^3)$. The randomized SVD further reduces cost to $O(nkp)$ for the leading k components. Numerical stability requires careful implementation, particularly for nearly degenerate spectra where small singular values may be computed inaccurately due to roundoff [4].

VIII. Connections to Related Methods

A. Factor Analysis

Factor Analysis (FA) [35] models the data as $\mathbf{x} = \mathbf{A}\mathbf{f} + \boldsymbol{\epsilon}$, where $\mathbf{f}$ is a low-dimensional latent factor vector, $\mathbf{A}$ is the loading matrix, and $\boldsymbol{\epsilon}$ is idiosyncratic noise with a diagonal covariance. Unlike PCA, FA explicitly separates shared variance (communality) from unique variance, and the loadings are estimated by maximum likelihood rather than eigen-decomposition. PCA can be viewed as FA with isotropic noise $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$, a model formalized as Probabilistic PCA [26].

B. Canonical Correlation Analysis

Canonical Correlation Analysis (CCA) [31] finds linear projections of two variable sets $\mathbf{X}$ and $\mathbf{Y}$ that maximize their mutual correlation. The canonical directions are obtained via SVD of the cross-covariance matrix $\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}^{-1/2} \boldsymbol{\Sigma}_{XY} \boldsymbol{\Sigma}_{YY}^{-1/2}$, making SVD the computational backbone of CCA as well.

C. Partial Least Squares

Partial Least Squares (PLS) [32] finds successive directions $\mathbf{w}$ and $\mathbf{c}$ in feature and response spaces that maximize the covariance $\text{Cov}(\mathbf{X}\mathbf{w}, \mathbf{Y}\mathbf{c})$. The first PLS direction is obtained via the left and right singular vectors of $\mathbf{X}^\top \mathbf{Y}$, again an SVD. PLS regression is preferred over PCR when the leading principal components of $\mathbf{X}$ are not predictive of $\mathbf{Y}$.

D. Non-Negative Matrix Factorization

Non-Negative Matrix Factorization (NMF) [33] decomposes $\mathbf{A} \approx \mathbf{W}\mathbf{H}$ with $\mathbf{W}, \mathbf{H} \geq 0$. Unlike SVD, which allows negative entries and negative cancellations, NMF produces a parts-based representation that is often more interpretable for data such as face images or document-term matrices. NMF minimizes $\|\mathbf{A} - \mathbf{W}\mathbf{H}\|_F^2$ subject to non-negativity, a non-convex problem solved by multiplicative update rules.

E. Kernel PCA

Kernel PCA [24] performs PCA in a high-dimensional feature space $\mathcal{H}$ implicitly defined by a kernel function $k(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle_{\mathcal{H}}$. By the kernel trick, the $n \times n$ kernel matrix $\mathbf{K}_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$ is eigen-decomposed instead of the $p \times p$ covariance matrix, enabling nonlinear dimensionality reduction without explicitly mapping to $\mathcal{H}$.

F. Independent Component Analysis

ICA [25] seeks a linear transformation $\mathbf{x} = \mathbf{A}\mathbf{s}$ such that the source signals $\mathbf{s}$ are statistically independent (not merely uncorrelated). PCA is typically used as a preprocessing step for ICA (whitening the data), and ICA can be seen as rotating the PCA components to achieve independence by maximizing non-Gaussianity (measured by negentropy or kurtosis). ICA finds application in blind source separation, EEG artifact removal, and financial returns analysis.

G. Dynamic Mode Decomposition

Dynamic Mode Decomposition (DMD) [34] analyzes time-evolving data by computing the leading eigendecomposition of the linear operator $\mathbf{A}$ mapping $\mathbf{x}_t \rightarrow \mathbf{x}_{t+1}$, estimated via SVD: $\mathbf{A} \approx \mathbf{X}'\mathbf{V}_k \boldsymbol{\Sigma}_k^{-1} \mathbf{U}_k^\top$. DMD modes are spatiotemporal coherent structures; their eigenvalues encode growth rates and oscillation frequencies. DMD is widely used in fluid dynamics and has found application in epidemiology and neuroscience.

IX. Conclusion

This survey has provided a comprehensive treatment of Principal Component Analysis and Singular Value Decomposition, tracing their mathematical foundations, algorithmic implementations, and broad applicability in data science. The central theoretical result—that PCA is optimally computed via SVD, and that truncated SVD provides the best low-rank approximation in both Frobenius and spectral norms (Eckart–Young–Mirsky)—underlies the practical success of these methods across domains ranging from genomics to collaborative filtering.

Both methods address all four themes of advanced predictive modeling: they expose linear dependencies in structured data through the covariance and singular value structure; they provide a principled basis for model construction by decorrelating inputs; they support forecasting via dimensionality-reduced regression and factor models; and they reveal latent structure in the form of principal components, singular modes, and low-dimensional embeddings.

At the same time, their limitations are important to appreciate: linearity constrains their applicability to non-Gaussian manifolds; scale sensitivity requires careful preprocessing; and interpretability of components is not guaranteed. The rich ecosystem of related methods—kernel PCA, ICA, NMF, CCA, PLS, probabilistic PCA, DMD—extends these ideas to address these limitations,

and understanding PCA and SVD deeply provides the foundation for mastering this broader landscape.

As data dimensionality continues to grow, driven by genomics, imaging, language models, and sensor networks, the fundamental geometric insight of PCA and SVD—that complex data often has low intrinsic dimensionality—remains one of the most powerful and enduring ideas in the field.

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